

|          |                                                                                                             |          |
|----------|-------------------------------------------------------------------------------------------------------------|----------|
| 2(a)(i)  | remove / cover, anthers (from plant <b>B</b> ) ;                                                            | <b>1</b> |
| 2(a)(ii) | method of transferring pollen (from anther) to stigma ;<br><b>or</b><br>shake anthers, on / over, stigmas ; | <b>1</b> |

| Question                             | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks               |                       |   |                       |                                     |     |    |       |                                   |     |    |       |                                      |    |    |       |                                  |    |    |         |            |  |  |          |   |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------|---|-----------------------|-------------------------------------|-----|----|-------|-----------------------------------|-----|----|-------|--------------------------------------|----|----|-------|----------------------------------|----|----|---------|------------|--|--|----------|---|
| 2(b)                                 | <p>any <b>three</b> from:</p> <p>light, intensity / duration / amount / strength / wavelength / colour ;</p> <p>temperature ;</p> <p>frequency of watering / amount of water ;</p> <p>humidity ;</p> <p>CO<sub>2</sub> / O<sub>2</sub> concentration ;</p> <p>growing medium / soil composition ;</p> <p>planting density ;</p>                                                                                                                                                                                                                               | 3                   |                       |   |                       |                                     |     |    |       |                                   |     |    |       |                                      |    |    |       |                                  |    |    |         |            |  |  |          |   |
| 2(c)(i)                              | <table><tr><th>offspring phenotype</th><th>O</th><th>E</th><th><math>\frac{(O - E)^2}{E}</math></th></tr><tr><td>purple petals<br/>long pollen grains</td><td>102</td><td>62</td><td>25.81</td></tr><tr><td>red petals<br/>round pollen grains</td><td>112</td><td>62</td><td>40.32</td></tr><tr><td>purple petals<br/>round pollen grains</td><td>14</td><td>62</td><td>37.16</td></tr><tr><td>red petals<br/>long pollen grains</td><td>20</td><td>62</td><td>28.45 ;</td></tr><tr><td colspan="3"><math>\chi^2 =</math></td><td>131.74 ;</td></tr></table> | offspring phenotype | O                     | E | $\frac{(O - E)^2}{E}$ | purple petals<br>long pollen grains | 102 | 62 | 25.81 | red petals<br>round pollen grains | 112 | 62 | 40.32 | purple petals<br>round pollen grains | 14 | 62 | 37.16 | red petals<br>long pollen grains | 20 | 62 | 28.45 ; | $\chi^2 =$ |  |  | 131.74 ; | 2 |
| offspring phenotype                  | O                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | E                   | $\frac{(O - E)^2}{E}$ |   |                       |                                     |     |    |       |                                   |     |    |       |                                      |    |    |       |                                  |    |    |         |            |  |  |          |   |
| purple petals<br>long pollen grains  | 102                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 62                  | 25.81                 |   |                       |                                     |     |    |       |                                   |     |    |       |                                      |    |    |       |                                  |    |    |         |            |  |  |          |   |
| red petals<br>round pollen grains    | 112                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 62                  | 40.32                 |   |                       |                                     |     |    |       |                                   |     |    |       |                                      |    |    |       |                                  |    |    |         |            |  |  |          |   |
| purple petals<br>round pollen grains | 14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 62                  | 37.16                 |   |                       |                                     |     |    |       |                                   |     |    |       |                                      |    |    |       |                                  |    |    |         |            |  |  |          |   |
| red petals<br>long pollen grains     | 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 62                  | 28.45 ;               |   |                       |                                     |     |    |       |                                   |     |    |       |                                      |    |    |       |                                  |    |    |         |            |  |  |          |   |
| $\chi^2 =$                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                     | 131.74 ;              |   |                       |                                     |     |    |       |                                   |     |    |       |                                      |    |    |       |                                  |    |    |         |            |  |  |          |   |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 2(c)(ii) | <p>1 reject <u>and</u> state that their calculated value / <math>\chi^2</math>, is higher than 7.82 / critical value ;</p> <p>any <b>two</b> from:</p> <p>2 (if the null hypothesis is rejected) There is a significant difference between the expected and observed values<br/><b>or</b><br/>there is a <b>less</b> than 5% / 0.05 probability that the difference has occurred due to chance ;</p> <p>3 genes, are on the same chromosome / are not on different chromosomes / are linked / do not assort independently<br/><b>or</b><br/>autosomal linkage<br/><b>or</b><br/><u>alleles</u> (likely to be) inherited together ;</p> <p>4 the two genes are close together (on the same chromosome) <b>or</b> small number of recombinants ;</p>                                               | 3     |
| 2(d)     | <p>1 yes in copper sensitive plants <u>and</u> no in copper tolerant plants / only in copper sensitive plants ;</p> <p>2 decrease, in <u>percentage / %</u>, viability, in sensitive plants / ora<br/><b>or</b><br/>slight increase, in <u>percentage / %</u>, viability, in tolerant plants ;</p> <p>3 in sensitive plants, there is no overlap of (error) bars, which means the <u>difference</u> may be significant<br/><b>or</b><br/>in tolerant plants, there is an overlap of (error) bars, which means <u>difference</u> is not significant ;</p> <p>4 no statistical test was carried out (in copper-sensitive plants) ;</p> <p>5 growth of pollen tube does not indicate pollen grain is viable ;</p> <p>6 no reference to concentration of copper ions used in the investigation ;</p> | 2     |